Notes: Gorton and Ordoñez (AER, 2014)
Collateral Crises

Contents

1 Big picture

This paper builds a theory of financial crises in which the key state variable is not simply leverage, but the **amount of information that lenders have about collateral**. The central object is short-term collateralized debt, interpreted as private money such as repo and other money-market instruments. This debt is valuable precisely because it is usually **informationally insensitive**: lenders are willing to roll over debt without producing costly information about the collateral.

The core mechanism is:

- In normal times, debt backed by opaque collateral can circulate without costly due diligence.
- Because lenders do not inspect collateral every period, public knowledge about who has good collateral and who has bad collateral gradually decays.
- This information decay allows more borrowers to obtain credit, including borrowers whose true collateral quality is low. Output rises during the boom.
- A sufficiently adverse shock can suddenly make information production privately attractive. Then debt becomes information-sensitive, collateral is revalued, borrowing contracts, and output falls.

Interpretation: The paper explains how a relatively small aggregate shock can trigger a large crisis after a long credit boom. The crisis is not caused by irrational exuberance or by multiple equilibria. It is caused by an endogenous transition from a regime in which collateral is not scrutinized to one in which collateral is scrutinized.

2 Incremental contribution of the paper

2.1 Conceptual contribution

The paper reframes financial crises as **collateral-information crises**. A crisis is a moment when private money stops being information-insensitive. This gives a precise interpretation of phrases such as "loss of confidence": lenders suddenly have incentives to investigate the collateral behind short-term debt.

2.2 Theoretical contribution

The paper embeds the idea of informationally insensitive debt into a dynamic macro model. The model jointly explains:

- why short-term collateralized debt is useful in normal times;
- why credit booms can be rational and output-enhancing;
- why booms endogenously create fragility;
- why a shock of the same size may have no effect early in a boom but cause a crisis later;
- why recovery can be slow when information has deteriorated for a long time.

2.3 Policy contribution

The paper shows that a social planner would typically produce **more information** than private agents, because private agents do not internalize how today's opacity affects future fragility and recovery. But the planner does **not** always want to eliminate fragility. Opacity has benefits because it allows credit to flow without costly information production.

Incremental message: The paper does not simply say "more transparency is always better." It argues that opacity can be socially useful, but excessive opacity creates dynamic fragility. The policy problem is therefore to choose how much information production to encourage, not to eliminate informationally insensitive debt altogether.

3 Economic environment

3.1 Agents and goods

The economy has two types of agents:

- **Households**, who begin with numeraire goods and can lend.
- **Firms**, who have managerial skills and collateral but need numeraire to produce.

There are two goods:

- **Numeraire** K, which can be used as productive capital.
- **Land**, a stand-in for collateral such as mortgage-backed securities or other asset-backed securities.

Firms have access to a stochastic Leontief production technology:

$$K' = \begin{cases} A\min\{K, L^*\}, & \text{with probability } q, \\ 0, & \text{with probability } 1-q. \end{cases}$$

Production is efficient if $qA > 1$, and the optimal production scale is $K^* = L^*$.

3.2 Collateral quality

Each firm owns one unit of collateral. Collateral can be:

- **Good**: pays C at the end of the period.
- **Bad**: pays zero.

Let p denote the common belief that a particular unit of collateral is good. Learning the true type requires paying an information cost γ.

Interpretation: Collateral is useful because firms cannot pledge output directly, but they can pledge land. Borrowing capacity therefore depends on perceived collateral quality and on whether lenders want to investigate the collateral.

4 One-period contracting problem

4.1 Information-sensitive debt

With information-sensitive debt, the lender pays γ to learn whether collateral is good. Lending then occurs only when the collateral is actually good. The expected profit from this contract is:

$$E(\pi \mid p, IS) = pK^*(qA - 1) - \gamma.$$

The benefit is that the lender avoids bad collateral. The cost is that every lender must spend resources on information production.

4.2 Information-insensitive debt

With information-insensitive debt, the lender does not inspect the collateral. The firm chooses a loan size small enough that the lender does not want to acquire information. The no-information condition is:

$$(1-p)(1-q)K < \gamma.$$

The feasible loan size under information-insensitive debt is:

$$K(p \mid II) = \min\left\{K^*, \frac{\gamma}{(1-p)(1-q)}, pC\right\}.$$

Expected profit is:

$$E(\pi \mid p, II) = K(p \mid II)(qA - 1).$$

Interpretation: If beliefs about collateral are high enough, or if information is sufficiently costly, firms can borrow without triggering due diligence. This is exactly the logic of private money: it works best when nobody needs to ask too many questions.

4.3 Choice between the two debt regimes

Firms induce information acquisition only if:

$$\frac{\gamma}{qA - 1} < pK^* - K(p \mid II).$$

Otherwise, they prefer information-insensitive debt.

The choice is non-monotone in the belief p:

- Very low p: information is not worth producing because collateral is probably bad.
- Intermediate p: information is valuable because learning separates good from bad collateral.
- High p: information is not worth producing because collateral is probably good.

Economic message: The most fragile region is the intermediate region where beliefs are high enough that collateral matters, but low enough that lenders may want to investigate. A shock can move the economy into this region.

5 Dynamics: information decay, credit booms, and crises

5.1 Information decay

The dynamic model uses overlapping generations. Households lend when young and become firms when old. Land can be transferred across generations and used as collateral.

Collateral quality changes over time through idiosyncratic shocks. With probability $1 - \lambda$, a unit of land is reshuffled and becomes good with probability $\hat{p}$. If no one produces information, beliefs drift toward the average $\hat{p}$.

Interpretation: Over time, without inspection, collateral histories become blurred. The economy forgets which collateral is good and which is bad. This is the paper's formal version of financial-market amnesia.

5.2 Credit boom

When beliefs converge to a sufficiently high average level, more collateral becomes eligible for borrowing. Firms with bad true collateral can borrow because lenders do not know which collateral is bad. Output rises.

This boom is not irrational. It is efficient ex post not to inspect every piece of collateral every period, because information production is costly.

5.3 Crisis

A negative aggregate shock lowers the perceived quality of collateral. If the shock is large enough relative to the current information state, lenders begin producing information, or firms reduce borrowing to avoid triggering that information production. Either way, credit contracts.

The same shock can have different effects depending on how long the previous boom lasted:

- Early in a boom, much collateral quality is still known, so the shock has limited effects.
- Later in a boom, information has decayed, so the shock affects a much larger stock of opaque collateral.

Economic message: The shock is exogenous, but the size of the crisis is endogenous. A long period of informationally insensitive lending makes the economy more fragile.

6 Policy implications

6.1 Why private agents underproduce information

Private agents are short-lived in the overlapping-generations model. They do not internalize how information production today affects future beliefs, borrowing, and recovery. This creates a wedge between private and social incentives.

A social planner values future generations and therefore treats information as an asset that improves future allocations. The planner's effective cost of information is lower because information today has future benefits.

6.2 Planner's trade-off

The planner produces more information than private agents, but not always enough to eliminate fragility. This is because informationally insensitive debt has real benefits:

- it economizes on costly monitoring;
- it allows broad collateral use;
- it sustains credit and output during normal times.

Policy interpretation: The planner should not necessarily "take away the punch bowl." The optimal intervention may be to reduce the amount of hidden fragility by encouraging more information production, while preserving the benefits of private money.

7 Numerical illustration

The paper illustrates the model with parameters:

$$1-\lambda = 0.1, \quad \hat{p} = 0.92, \quad q = 0.6, \quad A = 3, \quad K^* = L^* = 7, \quad C = 15, \quad \gamma = 0.35.$$

Information costs are therefore about five percent of the optimal loan. With these values, the important cutoffs are:

$$p^H = 0.88, \qquad p^L_{II} = 0.06, \qquad \text{information-sensitive region } p \in [0.22, 0.84].$$

The simulations introduce negative aggregate shocks in periods 5 and 50. The same shock is more damaging in period 50 because the economy has experienced a longer credit boom and more information has decayed.

Interpretation: The numerical section turns the model's verbal mechanism into a dynamic picture: credit booms raise production, but they also compress beliefs and make collateral more opaque. A later shock therefore causes a larger and more persistent decline.

8 Figures and reading guide

8.1 Figures 1–2: static information regimes and borrowing capacity

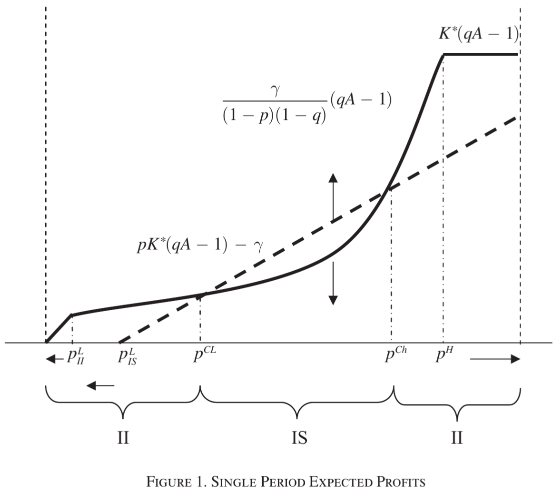

Figure 1. Single-period expected profits

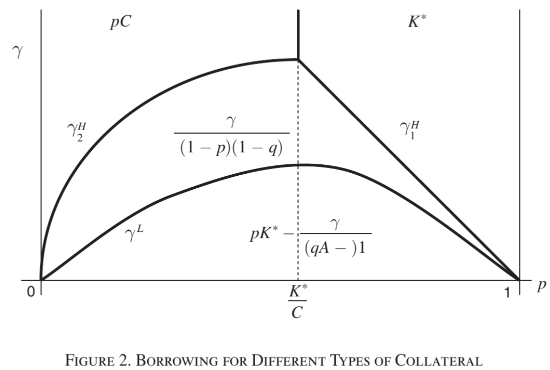

Figure 2. Borrowing for different types of collateral

Read:

- Figure 1 compares expected profits under information-sensitive debt and information-insensitive debt as beliefs p vary.
- The middle interval is the information-sensitive region: lenders inspect collateral because doing so is valuable.
- Low and high belief regions are information-insensitive: at low p, inspection is not worthwhile; at high p, collateral is sufficiently trusted.
- Figure 2 maps loan capacity across belief quality p and information cost γ.
- Full borrowing K^* is easiest when collateral has high perceived quality and information is costly to produce.

Economic message: Informational opacity can be privately and socially useful. Complicated collateral is attractive because it is harder to inspect, which can sustain credit. But this same property creates the possibility of abrupt regime changes when inspection becomes worthwhile.

8.2 Figures 3–4: numerical cutoffs and belief dynamics

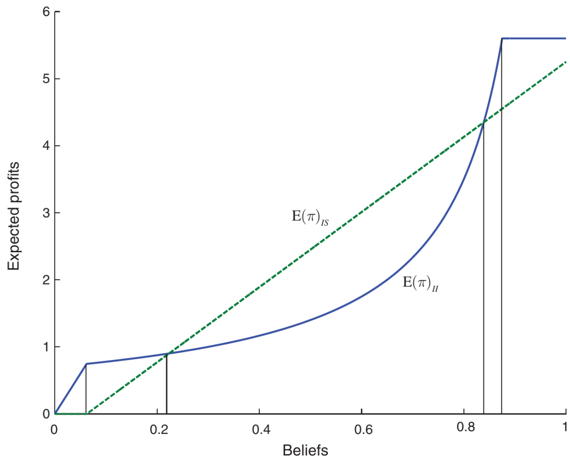

FIGURE 3. EXPECTED PROFITS AND CUTOFFS

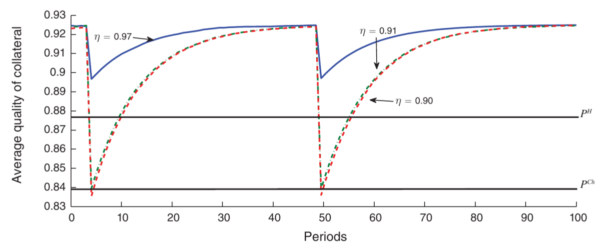

FIGURE 4. AVERAGE QUALITY OF COLLATERAL

Figure 3. Expected profits and cutoffs

Figure 4. Average quality of collateral

Read:

- Figure 3 calibrates the static cutoff logic from Figure 1 using the paper's numerical parameters.
- The green dashed schedule is information-sensitive expected profit; the blue curve is information-insensitive expected profit.
- Figure 4 shows how average perceived collateral quality evolves after shocks of size $\eta = 0.97$, 0.91, and 0.90.
- The shock lowers average collateral quality immediately, but mean reversion gradually pushes beliefs back toward $\hat{p} = 0.92$.

Economic message: The shock alone is not the whole story. Its effect depends on where it moves the economy relative to the information-sensitive cutoffs. A small shock can be benign in one state and destabilizing in another.

8.3 Figures 5–6: output collapse and information dispersion

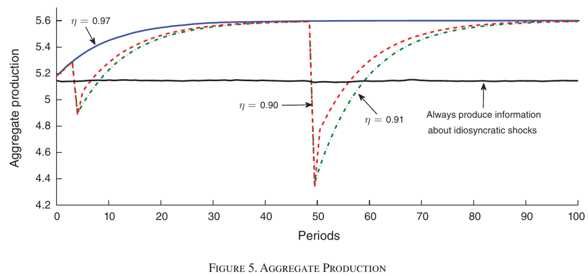

Figure 5. Aggregate production

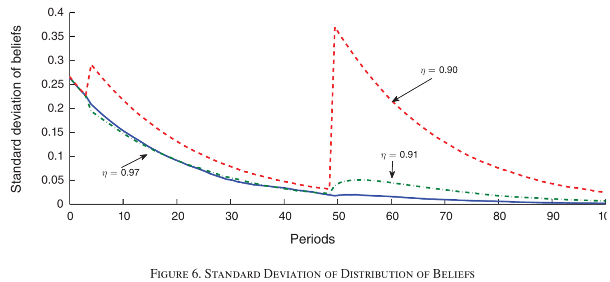

Figure 6. Standard deviation of distribution of beliefs

Read:

- Figure 5 plots output after shocks in periods 5 and 50.
- The same type of shock causes a larger fall in period 50 because the credit boom has had more time to erase information about collateral types.
- The horizontal black line shows the case where information about idiosyncratic shocks is always produced. Output is lower in normal times but less fragile.
- Figure 6 shows dispersion of beliefs. During a boom, dispersion falls because collateral looks increasingly similar.
- A crisis can increase dispersion sharply because inspection reveals which collateral is good and which is bad.

Economic message: A boom is not just high credit. It is also low information dispersion. When collateral becomes gray rather than black or white, more lending occurs, but the system becomes more vulnerable to sudden reclassification.

8.4 Figures 7–8: extreme information costs and the planner's information choice

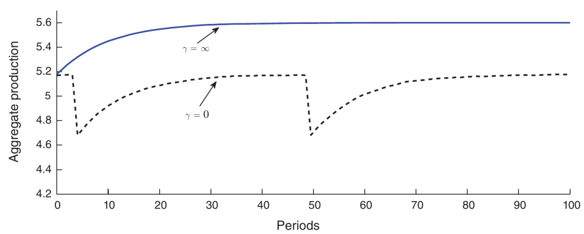

FIGURE 7. EXTREME INFORMATION COSTS

Figure 7. Extreme information costs

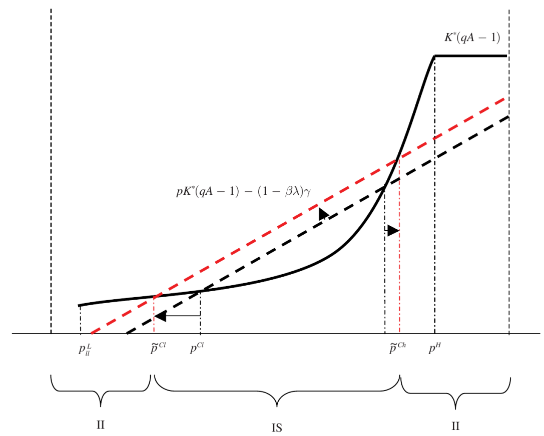

FIGURE 8. INFORMATION ACQUISITION BY THE PLANNER

Figure 8. Information acquisition by the planner

Read:

- Figure 7 compares two extreme cases: information is free ($\gamma = 0$) and information is impossible ($\gamma = \infty$).
- When information is free, output is lower and more volatile because only good collateral can sustain borrowing.
- When information is impossible, all collateral remains pooled and production is smoother.
- Figure 8 compares decentralized information acquisition with the planner's decision.
- The planner's information-sensitive region is wider because information today improves future allocations.

Economic message: Perfect transparency is not automatically best. The planner values information more than private agents do, but also recognizes that too much information production can destroy the liquidity benefits of private money.

8.5 Figure 9: decentralized economy versus planner after a shock

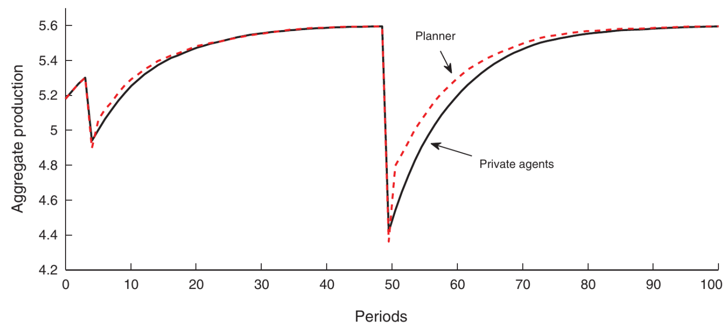

FIGURE 9. DYNAMICS WITH AN AGGREGATE SHOCK $\eta = 0.91$

Figure 9. Dynamics with an aggregate shock $\eta = 0.91$

Read:

- The solid curve represents private agents; the dashed curve represents the planner.
- At the moment of the shock, the planner may accept lower current production by producing more information.
- After the shock, the planner recovers faster because information has been replenished.

Economic message: The planner trades off current credit against future fragility. Private agents prefer to keep borrowing in the crisis year and ignore the slower recovery they impose on future generations.

9 Main takeaways

1. **Private money works because it is informationally insensitive.** Short-term collateralized debt is useful when lenders can accept collateral without costly investigation.
2. **Credit booms can be rational.** A boom emerges because not producing information allows more firms to borrow and output to rise.
3. **Fragility is endogenous.** The longer the economy goes without inspecting collateral, the more information decays, and the more fragile the system becomes.
4. **Crises are regime switches.** A crisis occurs when lenders suddenly have incentives to produce information or borrowers cut back to prevent that information production.
5. **Opaque collateral is endogenous.** Agents prefer collateral that has high perceived quality and high information-acquisition costs; this helps explain why complex mortgage-backed securities became important in the recent crisis.
6. **Policy is not simply transparency versus opacity.** The planner wants more information than private agents produce, but does not necessarily want to eliminate all fragility.

10 How to position this paper in the literature

10.1 Compared with leverage-amplification models

Traditional collateral amplification models emphasize fire sales and falling collateral prices. This paper instead emphasizes **information production**. The crisis is not only that collateral prices fall; it is that collateral becomes information-sensitive.

10.2 Compared with multiple-equilibrium bank-run models

Diamond-Dybvig style models often rely on self-fulfilling runs. This paper generates crises without equilibrium multiplicity. The trigger is a unique endogenous switch in the value of producing information.

10.3 Compared with behavioral credit-boom stories

The boom is not caused by irrational optimism. Private agents choose opacity because information is costly. The problem is dynamic: their choices do not internalize future fragility.

11 Short summary

Gorton and Ordoñez provide a theory in which financial crises arise from the dynamics of information about collateral. In normal times, short-term collateralized debt works well because nobody needs to inspect the collateral. Over time, this lack of inspection allows collateral quality information to decay. The resulting opacity expands credit and output, but it also creates fragility. When a negative shock makes information production attractive, collateralized debt stops being private money and becomes information-sensitive. Credit contracts and output falls. A planner would produce more information than private agents, but would still preserve some informational insensitivity because private money has real benefits.

12 Conclusion

The paper's main insight is that crises are not just about bad shocks; they are about the informational state of the financial system when shocks arrive. A small shock can be harmless when collateral quality is well understood, but destructive after a long boom has eroded that information. The paper therefore gives a disciplined explanation for why credit booms often precede crises and why complex, opaque collateral can be both useful and dangerous.